Multivariate Analysis of the Physicochemical Characteristics of Portuguese Vinho Verde Wines

Introduction

Wine quality has historically been evaluated through subjective sensory assessments. However, the objective physicochemical characteristics of wine—such as acidity, residual sugar, alcohol, and sulfur dioxide—are the fundamental building blocks that shape its profile and preserve its shelf life. Understanding the complex, multidimensional relationships among these chemical properties is essential for viticulturists and data analysts alike. This study utilizes multivariate statistical techniques to investigate how these chemical properties differentiate wine types (red vs. white), how their dimensionality can be reduced, and what underlying latent factors drive wine chemistry.

Dataset

The data used for this analysis is the Wine Quality Dataset, sourced from the UCI Machine Learning Repository. It contains physicochemical measurements of Portuguese Vinho Verde wines. The combined dataset includes 6,497 total observations, comprising 1,599 red wine samples and 4,898 white wine samples. The dataset contains 11 continuous physicochemical variables (fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, density, pH, sulphates, and alcohol) and a categorical variable identifying the wine color (color). The sensory quality score was excluded from the multivariate equations, as the focus is solely on the objective physicochemical characteristics.

Research Questions and Hypotheses

This study addresses three primary research questions:

- **RQ1 (MANOVA):** Do red and white wines differ significantly in their combined physicochemical characteristics?
    - H_0: The mean vector of the selected physicochemical variables is the same for red and white wines ($\mu_{red} = \mu_{white}$).
    - H_1: The mean vectors differ between red and white wines ($\mu_{red} \neq \mu_{white}$).
- **RQ2 (PCA):** Can the 11 physicochemical measurements of wine be summarized using a smaller number of principal components while retaining a substantial proportion of the total variability?
- **RQ3 (EFA):** What underlying latent dimensions or factors explain the relationships among the physicochemical characteristics of wine?

Methodology

The analysis was conducted using Python via Jupyter Notebook (referencing "APM1220_SA1_SANTIAGO_TUPAS.ipynb - Colab.pdf"). The pandas, numpy, matplotlib, seaborn, scipy, statsmodels, scikit-learn, and factor_analyzer libraries were utilized.

- **Data Preparation & EDA:** The red and white wine datasets were concatenated, and exploratory data analysis was conducted to check for missing values, duplicates, and outliers.
- **MANOVA:** A Multivariate Analysis of Variance was performed to test for group differences using four distinct variables: volatile acidity, total sulfur dioxide, residual sugar, and chlorides. Pillai's Trace was prioritized due to violations of covariance homogeneity (Box's M test).
- **PCA:** Principal Component Analysis was applied to standardized data (z-scores) to eliminate scale disparities and reduce dimensionality.
- **EFA:** Exploratory Factor Analysis was initiated to identify latent constructs.

Results

1. Exploratory Data Analysis

The combined dataset contained no missing values but included 1,177 duplicate rows. Mahalanobis distance identified 107 multivariate outliers at $p < 0.001$, representing legitimate variations in specialty wines rather than errors. Total and free sulfur dioxide exhibited the largest variances. White wines contained significantly higher concentrations of sulfur dioxide to prevent oxidation, while red wines exhibited higher volatile acidity, chlorides, and sulphates.

2. MANOVA

Four non-redundant variables were selected: volatile acidity, total sulfur dioxide, residual sugar, and chlorides. Box's M test was significant ($p = 0.000$), so Pillai's Trace was utilized for robustness. The MANOVA results indicate a statistically significant difference (Pillai's trace = 0.6983, $F = 3756.05$, $p < 0.001$). Follow-up ANOVAs revealed massive effect sizes for total sulfur dioxide ($\eta^2 = 0.4905$) and volatile acidity ($\eta^2 = 0.4265$), confirming that red and white wines possess fundamentally distinct multivariate chemical signatures.

3. PCA

The 11 physicochemical variables were standardized to a mean of ~0 and standard deviation of ~1. Based on the Kaiser criterion (eigenvalues > 1.0) and the visual "elbow" of the Scree plot, 4 principal components were retained, capturing 73.19% of the total cumulative variance.

- **PC1 (Physical Body & Alcoholic Profile):** Driven by density, alcohol, and residual sugar.
- **PC2 (Sulfur Dioxide & Preservation Factor):** Driven by total and free sulfur dioxide.
- **PC3 (Acid Balance & Tartness Profile):** Driven by fixed acidity, pH, and citric acid.

- **PC4 (Mineral & Volatile Impurity Dimension):** Driven by chlorides and volatile acidity.

4. EFA

Bartlett's Test of Sphericity yielded a significant chi-square of 37094.62 ($p = 0.000$), indicating sufficient correlation among variables. However, the Kaiser-Meyer-Olkin (KMO) measure of sampling adequacy returned a value of 0.4053. While the data technically falls below the standard >0.60 threshold for robust factorability, the preliminary eigenvalues and Scree plot suggest the presence of 3 to 4 extractable latent factors mapping to preservation practices, acidic structure, and fermentation profiles.

Integrated Discussion

1. **MANOVA: What distinguishes red and white wines?** Red and white wines display a statistically significant overall difference in their combined physicochemical characteristics. Red wines are primarily distinguished by higher volatile acidity and chlorides, whereas white wines are heavily defined by significantly higher total and free sulfur dioxide concentrations used for preservation.
2. **PCA: What are the major dimensions of variation?** The PCA demonstrates that the 11 physicochemical measurements can be successfully reduced to four major components that capture 73.19% of the variance. These dimensions represent the primary axes of physical variation, contrasting the wine's physical body/density (PC1), sulfur preservation tactics (PC2), natural grape acidity (PC3), and mineral impurities (PC4).
3. **EFA: What latent factors explain the relationships?**
   The preliminary EFA models the *common variance* to uncover theoretically interpretable, latent enological practices. These underlying factors cleanly map to broad winemaking decisions: "Sulfur Management" (preservation), "Natural Grape Acidity" (tartness and pH balance), and "Fermentation Outcomes" (residual sugar and alcohol content).
4. **Comparison of PCA and EFA:**
   Both PCA and EFA highlight sulfur dioxide, acidity, and density as critical structural variables. The primary difference is methodological: PCA maximizes *total* variance explained (resulting in principal components that mathematically force orthogonal dimensions, like blending density and sugar into PC1), while EFA models the *shared* variance to uncover correlated latent factors that reflect real-world, interconnected winemaking decisions (e.g., separating preservation additives from natural grape acids).
5. **Substantive interpretation:** The integrated findings confirm that the physicochemical characteristics of Vinho Verde wines are not random but are strictly dictated by the wine type and specific fermentation decisions. The chemical makeup is fundamentally structured around three to four core dimensions: how the wine is preserved (sulfur), its natural grape characteristics (acidity/pH), its fermentation profile (sugar/alcohol/density), and its mineral/spoilage limits (chlorides/volatile acidity).
6. **Limitations:**

- **Imbalanced Data -** The dataset contains a severe imbalance, 4,898 white wines vs. 1,599 red wines, which makes the multivariate centroids, principal components, and factor structures are disproportionately influenced by the characteristics of white wines.
- **Limitations of Applying PCA/EFA to Chemical Variables -** Physicochemical wine characteristics might interact in complex, non-linear ways during fermentation and aging, which might make linear reduction techniques oversimplify the true chemical profile of the wines.

Conclusion

This multivariate analysis demonstrates that Portuguese Vinho Verde wines can be systematically modeled using their physicochemical properties. MANOVA confirmed that red and white wines possess significantly different multivariate chemical profiles, driven heavily by sulfur dioxide and volatile acidity. PCA effectively reduced the high-dimensional chemical data into four principal components, capturing over 73% of the variance by isolating body, preservation, acidity, and impurities. While the EFA faced limitations with sampling adequacy (KMO = 0.4053), the overall integrated models reveal that wine chemistry is fundamentally structured around targeted enological decisions regarding preservation, acidity, and fermentation.

References

Cortez, P., Cerdeira, A., Almeida, F., Matos, T., & Reis, J. (2009). *Modeling wine preferences by data mining from physicochemical properties*. Decision Support Systems, 47(4), 547-553.

UCI Machine Learning Repository: *Wine Quality Dataset* (https://archive.ics.uci.edu/dataset/186/wine+quality). Licensed under CC BY 4.0.